HACKPIRE LUMINA

TEAM NAME: **Hyperion** PROJECT TITLE: **Project Heimdall**

PROBLEM

- Malaysian commercial buildings pay TNB a **peak demand charge** set by their single highest 30-minute interval each month: RM89–97 per kW, about 3× higher since July 2025.
- Facility managers can't see peaks coming, so **one hot afternoon sets the whole month's bill**.
- Peaks also strain a national grid already stretched by the data-centre boom.

CUSTOMER ELEMENT

- **Mid-sized commercial buildings** on TNB medium-voltage Time-of-Use tariffs with central chillers: malls, office towers, hospitals, campuses, hotels.
- **Buyer:** building owners and property groups (one deal, many buildings).
- **Daily user:** facility managers, who are blamed for high bills.

UNIQUE VALUE PROPOSITION

- A forecast that comes with an action: **a pre-cooling plan sent the evening before**.
- Software only: no solar panels or batteries to buy.

We help facility managers of commercial buildings cut their monthly peak demand charge by forecasting tomorrow's load and planning when to pre-cool.

SOLUTION

- **Day-ahead load forecast** (best guess and cautious P95) from meter data, weather and calendar.
- **Peak plan:** if the cautious forecast breaks the monthly target during 2–10pm, shift chiller load into the off-peak morning.
- **Waste check:** flags usage 20%+ above forecast for 3+ hours.
- Web dashboard plus a daily WhatsApp plan.

CHANNELS

- **Free peak audit:** owners share 12 months of TNB interval data; we show what Heimdall would have saved.
- Direct outreach to property groups and facility managers.
- Partners: facility-management firms and solar or efficiency installers.
- Access: web dashboard and daily WhatsApp or email plan.

REVENUE/FUNDING MODEL

- **Monthly subscription** per building.
- **Or shared savings:** a cut of verified peak-charge savings. No savings, no fee.
- Later: green-technology grants, and selling pooled building flexibility to the grid as demand response.

COST STRUCTURE

- Cloud hosting and monthly model retraining.
- Developer time to maintain the model and dashboard.
- Meter data integration per building.
- Sales and onboarding.
- Software only, so the cost of each extra building is low.

KEY METRICS

- **Backtest, 99 buildings:** monthly peak 3.1% lower, about **RM1,140 per building per month**.
- Forecast error 6.1% vs 8.6% for the "same hour last week" shortcut.
- 93% of peak-raising hours caught by a plan.
- In pilots: kW cut, RM saved, plan follow-rate, buildings onboarded.

UNFAIR ADVANTAGES

- A **working, backtested model** on 99 real buildings, not a mock-up.
- Time-series forecasting and anomaly-detection experience from trading financial markets.
- Our insight: savings come from timing. TNB does not charge maximum demand that occurs off-peak, so pre-cooling before 2pm is safe.

ROADMAP

PHASE 1 DONE

- Two years of hourly meter data from 99 hot-climate commercial buildings (Building Data Genome 2), cleaned of meter glitches and joined with weather.
- TNB Time-of-Use rules verified.
- Cost: free open data and team time.

PHASE 2 DONE

- Gradient boosting (LightGBM) with quantile forecasts (P50, P95) for day-ahead load; residual-based anomaly detection for waste.
- Forecasts turned into a pre-cooling plan and a working dashboard prototype.

PHASE 3 NEXT

- Backtested on 9 unseen months (done); next, a pilot with one Malaysian building on its 30-minute TNB data.
- Impact measured as kW and RM saved vs the same months last year; then pool buildings for grid demand response.

TIPS: Keep your answers concise and specific. Focus on the real-world problem, the value of your solution, and how you will measure its impact. If a section is not applicable, write N/A and briefly explain why.